Chirality without firmness

Human Compression project

In the final published edition of Leemans–Tranchida [1], printed page 492, a geometry is an incidence system in which every flag extends to a chamber. Firmness is defined separately. On page 493, chirality requires exactly two orbits of the full type-preserving group on chambers, with adjacent chambers in different orbits. These definitions permit singleton panels. The following example shows why the omission matters for the unrestricted auxiliary statements.

1 An example with seven objects

Let the three type classes be

$$X_0 = \{d, c_1, c_2\}, \qquad X_1 = \{x_1, x_2\}, \qquad X_2 = \{y_1, y_2\}.$$

Apart from reflexivity, declare precisely the following pairs incident, for $j = 1, 2$:

$$x_j \sim y_j, \qquad d \sim x_j, \quad d \sim y_j, \quad c_j \sim x_j, \quad c_j \sim y_j. \tag{1}$$

Call the resulting system Γ.

Proposition 1. *The system Γ is a connected geometry satisfying the literal chirality definition of [1]. Its full type-preserving group is C_2. It is not firm. Moreover, all residues of corank at least two at the chamber $C_1 = \{c_1, x_1, y_1\}$ are connected, but the residue of d is disconnected.*

Proof. The only chambers are

$$C_j = \{c_j, x_j, y_j\}, \qquad D_j = \{d, x_j, y_j\} \quad (j = 1, 2).$$

Every pair in (1) lies in one of these chambers, and every object lies in a pair. Hence every flag extends to a chamber. The whole incidence graph is connected: d is joined to every x_j, y_j, and c_j is joined to x_j.

Among the type-zero objects, d has degree four and the other two have degree two. Every type-preserving automorphism fixes d. The matching $x_j \sim y_j$ forces the same permutation of the two indices on types one and two. The incidences of c_j then force that same permutation on c_1, c_2. Conversely, simultaneous exchange of the two indices preserves all incidences. Thus the full group is exactly C_2, with chamber orbits $\{C_1, C_2\}$ and $\{D_1, D_2\}$. The only adjacent chamber pairs are C_j, D_j, so adjacency always crosses between the two orbits. This is the stated chirality condition.

The panel $\{c_j, x_j\}$ has only the completion y_j, so the geometry is not firm. In fact eight panels have one completion and two panels have two completions.

The flags in C_1 of corank at least two are the empty flag and the three singleton flags. The empty residue is the connected whole geometry. The residue of c_1 is the single edge $x_1 \sim y_1$. The residue of x_1 has vertices c_1, d, y_1 and edges $c_1 \sim y_1 \sim d$; the residue of y_1 is the analogous path. All four residues are connected. In contrast, the residue of d has vertices x_1, x_2, y_1, y_2 and consists of the two disjoint edges $x_1 \sim y_1$ and $x_2 \sim y_2$. □

The example refutes three literal auxiliary assertions: the inference from chirality to thinness on printed page 493; Theorem 4.1 on page 494, applied to C_1; and Proposition 4.2 on page 495. For the last assertion, already the proper flag type $\{0\}$ has two object orbits, $\{d\}$ and $\{c_1, c_2\}$. The source proofs use an adjacent chamber where firmness has not been supplied. With firmness as an additional hypothesis, both normalization arguments are valid and are retained in the project proof bank.

This example is not a standard C^+ coset geometry. Indeed no subgroup of its full type-preserving group is transitive on the type-zero objects, whereas a coset action is transitive on each type. Thus it does not refute the corrected higher-rank variant of Corollary 4.9. The separate cyclic example handles the unrestricted wording of that corollary in rank two.

2 The rank of the residue in Proposition 4.7

The displayed statement of Proposition 4.7, printed page 496, requires $|J| < |I|$ and asserts connectedness of the residue under

$$G_J = \langle G_{J\cup\{i\}} : i \in I \setminus J \rangle. \tag{2}$$

The preceding discussion excludes rank-one residues as irrelevant to residual connectedness. This contextual restriction must also be present in the proposition: (2) does not imply connectedness in rank one. For example, in the already accepted rank-four C^+ geometry of [2], take any base panel with type set J, so $|J| = 3$. The intersection property gives $G_J = G_I = 1$, and hence (2) holds. But the residue consists of two objects of the same type. Its incidence graph has two isolated vertices and is disconnected. The existing recovery of this sufficient criterion is explicitly for residues of rank at least two and for the designated action with proper-flag transitivity. The source's separate switch between the full automorphism group and the designated group is documented in the accepted principal counterexample; neither qualification is silently removed here.

3 Verification and scope

The accompanying finite check enumerates every flag of the seven-object system and all $3!\,2!\,2! = 24$ permutations preserving its types. It finds exactly the two automorphisms proved above, the two chamber orbits, the complete panel counts and the four connected base residues. The disconnected residue of d is checked separately. No group search, classification or automorphism software is required.

The definitions and statements belong to [1]; the seven-object construction and proof are supplied here without a priority claim. The rank-one observation imports the reviewed data of [2]. These auxiliary corrections add no original, open-question solution, Lean theorem or compression measurement. The completion ledger keeps the literal assertions separate from their corrected variants.

References

[1] D. Leemans and P. Tranchida, *On residual connectedness in chiral geometries*, Algebraic Combinatorics **4** (2021), 491–499, doi:10.5802/alco.162.

[2] Human Compression project, *A chiral coset geometry with disconnected residues*, reviewed project manuscript, five pages; exact proof and review bound in the dependency ledger.